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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

PHY 300 — Past Questions & Materials

by Renance

Course: PHY 300

Institution: Federal University of Petroleum Resources, Effurun

Papers in this pack: 4

Levels: 200, 300

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Paper 1 of 4 — PHY 300 - 2019/2020 - First Semester - Experimental Physics V

Level 300 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES,
EFFURUN
DEPARTMENT OF PHYSICS
FIRST SEMESTER EXAMINATION.

Course Code: PHY 300
Course Title: Experimental Physics V
Course Unit: One
Session: 2019/2020
Time Allowed: 1½ Hour
Instructions: ANSWER ALL QUESTIONS

Question 1

- (a) Explain the conversion of A.C voltage to D.C voltage using circuit diagram and wave form
- (b) (i) Mention three passive and two active electronics components
(ii) State the uses and symbols of these 5 electronics components mentioned in b(i) above

Question 2

Using block diagrams design and explain the following

- (a) Flower pot control and an irrigation system that irrigates only when it does not rain
- (b) Quiz timer and range control

Paper 2 of 4 — PHY 300 - 2016/2017 - Second Semester - Physics

Level 200 • Second Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES

COLLEGE OF SCIENCE

DEPARTMENT OF PHYSICS

SECOND SEMESTER EXAMINATION 2016/2017 ACADEMIC SESSION

COURSE CODE: PHY 300 TIME ALLOWED: 1 HOUR

INSTRUCTION: ANSWER ALL QUESTIONS

1. (a) State and discuss the main components in inverter that convert D.C voltage to A.C voltage with simple circuit diagram (10 marks)
(b) Explain the conversion of A.C voltage to D.C voltage using circuit diagram and waveform (5 marks).
2. (a) Design and explain using block diagrams the full Flower Pot Control an irrigation system, that irrigates only when it does not rain (10 marks).
«OM Boplain latching and memory cells-(5-marks) m
(ii) Implement “selfHolding” using ladder diagram and explain how the system latches itself ON (10 marks).

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Paper 3 of 4 — Phy 300 Exam PQ

Level 300 • First Semester • PastQuestion

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FIRST SEMESTER EXAMINATION.

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Paper 4 of 4 — PHY 300 2016 2017 EXPERIMENTAL PHYSICS 4

Level 300 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES

COLLEGE OF SCIENCE

DEPARTMENT OF PHYSICS

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(c) Implement "self-Holding" using ladder diagram and explain how the system latches itself ON (10 marks).

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